BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming: Files, SQL and Strings

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is a parameter?

2. Which is a benefit of a high-level language?
   A) Never needs translation
   B) Easier for humans to read
   C) Runs directly on every CPU
   D) Uses no programming rules

3. "Adding active users always increases the bandwidth available to each user." Correct this. Explain why.

4. A school needs central accounts and shared files. Three club laptops share files directly. Correct both claims. Explain why. (a) Every client must be a dedicated server. (b) Peer-to-peer needs a central server.

Today's objectives

Explain the core ideas of programming: files, sql and strings using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

String handling
Concatenate and extract parts of text when cleaning/formatting data.

File processing
Open, read/write and close files in the appropriate mode.

SQL retrieval
SELECT selects fields, FROM selects table and WHERE filters rows.

Combined tasks
A realistic system may read records, filter content and produce a formatted result.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Link the representations

The same three pupils can appear as CSV text lines, a 2D table and a typed SQL database. The representation changes, not the facts.

Separate indexing from filtering

A 2D array uses indexes for cell access; a loop and IF are needed to select every row satisfying a condition.

SQL query

SELECT chooses displayed fields, FROM names the table and WHERE selects matching records.

Optional integration

Loading CSV rows into an array and converting string marks into numbers is an ambitious extension; start with a supplied array for core work.

I Do - Worked model

Compare SQL and code for the shared Pupils table. Show why 'SELECT Name FROM Pupils WHERE Mark >= 50' returns Ada and Mia.

- Read marks 72, 43 and 81 from the Pupils table; identify that 50 is the minimum included mark.
- SQL SELECT Name requests only the Name field; WHERE Mark >= 50 keeps Ada and Mia.
- In code, loop through aligned names=['Ada','Ben','Mia'] and numeric marks=[72,43,81]; test each mark>=50.
- Print names[0] and names[2]. The code and SQL give the same set of names, although ordering is not guaranteed by SQL without ORDER BY.

We Do - Guided practice

1. Using the SAME dataset, write SQL and describe code that returns the Name of pupils with Mark < 60. [5 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write a SQL query returning all titles from Books where Available = 1.

```
SELECT Title FROM Books ...
```

You Do - Independent exam questions

Q2. Write SQL to display Name from Pupils for pupils whose Mark >= 70. [3 marks]

Q3. Complete a code condition so that names[i] is printed only when marks[i] < 50. State the result using the shared data. [3 marks]

Q4. Explain why comparing '72' directly with numeric 50 can be problematic when reading marks from a CSV text file. [3 marks]

Exit Ticket - True or False?

SQL SELECT identifies the table to search.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A named input received by a subprogram when called with an argument.

2. B - Easier for humans to read

3. Users may compete for finite bandwidth, reducing the data rate available to each.

4. (a) A server manages shared services and resources for clients. (b) Peers share directly and can act as clients and servers. Central control suits a school; peer-to-peer can cost less for a small club but is harder to manage as it grows.

I Do - Explain the model

Read marks 72, 43 and 81 from the Pupils table; identify that 50 is the minimum included mark.

SQL SELECT Name requests only the Name field; WHERE Mark >= 50 keeps Ada and Mia.

In code, loop through aligned names=['Ada','Ben','Mia'] and numeric marks=[72,43,81]; test each mark>=50.

Print names[0] and names[2]. The code and SQL give the same set of names, although ordering is not guaranteed by SQL without ORDER BY.

We Do - Model answers

1. Equivalent code loops through indexes 0-2 and prints the name only when marks[i] < 60; output Ben.

Code First - Answer

SELECT Title FROM Books WHERE Available = 1;
Expected output: titles of rows with Available = 1 (table supplied)

You Do - Self-mark

Q2: SELECT Name [1] FROM Pupils [1] WHERE Mark >= 70 [1]; Ada and Mia match.

Q3: IF marks[i] < 50 THEN [2 for correct condition/control] print(names[i]) [1]; prints Ben.

Q4: readLine() supplies text [1]; '72' is a string, while 50 is numeric [1]; the field must be converted/validated before a numerical comparison [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.